

System Requirements Specification (SRS)

Project Name: CareerVault

Version: 1.0.0

Architecture: Web 2.5 (Hybrid: SQL + Merkle Anchoring)

1. Introduction

1.1 Purpose

To build a **Career Verification Platform** that allows verified organizations to issue tamper-proof, blockchain-anchored career documents (LORs, Experience Letters) with zero technical friction.

1.2 Core Value Proposition

- **For Issuers:** "DocuSign-like" ease of use. No wallets, no keys.
- **For Holders:** Permanent ownership of career history, immune to company bankruptcy.
- **For Verifiers:** Instant, cryptographic verification of candidate claims.

1.3 The "Web 2.5" Philosophy

The system uses a standard **PostgreSQL database** for all user interactions (fast/cheap) and uses a **Daily Merkle Tree Anchor** on the **Polygon Blockchain** for immutable audit trails (secure/permanent).

(Note for Devs: We batch thousands of documents into one hash to keep gas costs near zero.)

2. Actors & User Roles

Role	Access Level	Description
1. Org Admin	High	Sets up the Company Profile, verifies Domain (DNS), manages the "Allowlist" of authorized signers.
2. Issuer (Manager)	Medium	The direct supervisor. can Draft and Sign LORs, but cannot <i>Finalize</i> them without HR approval.
3. Approver (HR)	High	The "Gatekeeper." Receives signed drafts from Managers. Cross-checks records. Finalizes the issuance.
4. Holder (User)	Personal	The Employee. Requests docs, manages their "Career Wallet," pays to generate "Shareable Links."
5. Verifier	Public/Paid	Recruiters or background check firms who view/verify the documents.

3. Functional Requirements

3.1 Module A: Organization Onboarding (Trust Layer)

- **REQ-A1 (Domain Verification):** System must generate a unique TXT token (e.g., cv-verify-123). Admin must place this in their Company DNS records. System scans DNS to verify ownership of @company.com.
- **REQ-A2 (The Allowlist):** Admin must upload/manage a list of **Authorized Manager Emails** (or allow *@company.com with a warning).
- **REQ-A3 (HR Assignment):** Admin must designate specific users as **Approvers (HR)**.

3.2 Module B: The 2-Step Issuance Flow (The Loophole Fix)

- **REQ-B1 (Step 1: Manager Signature):**
 - Manager receives a Magic Link request from Employee.
 - Manager reviews draft -> Clicks **"Sign & Submit to HR."**
 - *Status:* PENDING_HR_APPROVAL.
- **REQ-B2 (Step 2: HR Validation):**
 - HR receives notification: *"Manager Mark signed an LOR for Sarah. Please Verify."*
 - HR checks internal records (e.g., dates of employment).
 - HR clicks **"Finalize Issuance."**
 - *Status:* ISSUED.
- **REQ-B3 (Rejection Loop):** If HR rejects, they must provide a reason ("Wrong Dates," "Not Eligible"). Manager and Employee are notified.

3.3 Module C: Document Generation (Standard Template)

- **REQ-C1 (JSON-LD):** System generates a W3C-compliant JSON-LD file containing:
 - Issuer DID (The Company)
 - Signer ID (The Manager's Name/Email)
 - Approver ID (The HR's Name/Email)
 - Holder ID (Sarah)
 - Content (Role, Dates, Rating, Text)
 - Extracted Skills (Array of AI-mapped competencies)
- **REQ-C2 (PDF Rendering):** System generates a **Standardized PDF** (Clean, Professional, System-Branded) and embeds the JSON-LD inside the PDF metadata.

3.4 Module D: The "Anchor" Engine (Web 2.5 Core)

- **REQ-D1 (Hashing):** System calculates SHA-256(Salt + JSON).
- **REQ-D2 (The Cron Job):** Every 24 hours (00:00 UTC), System collects all ISSUED hashes from the last day.
- **REQ-D3 (Merkle Tree):** System builds a Merkle Tree from these hashes and obtains the Root Hash.
- **REQ-D4 (Blockchain Write):** System writes **only the Root Hash** to the Polygon Smart Contract.
- **REQ-D5 (Proof Generation):** System generates a "Merkle Proof" for each specific document and saves it to the DB.

3.5 Module E: Verification & Monetization

- **REQ-E1 (The Paywall - Holder Side):**
 - Sarah can see her LORs for free in her private dashboard.
 - To share one, she clicks "Generate Verified Link."
 - **Trigger:** System prompts for payment (Subscription or One-time fee) -> Generates public URL.
- **REQ-E2 (The Paywall - Verifier Side):**
 - If a Company (Verifier) wants to verify a *bulk* set of candidates or access API verification, they must pay.
 - **Discount Logic:** If the Verifier Company is *also* an Onboarded Issuer (Module A), they get a 50% discount on verification costs.

3.6 Module F: AI-Powered Analysis & Talent Matchmaking (The Smart Layer)

- **REQ-F1 (Semantic Skill Extraction - Enhancing Module C):**
 - **Trigger:** When a Manager submits an LOR draft.
 - **Action:** An ML microservice (e.g., using SpaCy or a local LLM) parses the unstructured text of the LOR.
 - **Output:** The system automatically extracts core competencies (e.g., "React," "Agile Management," "System Design") and maps them to a standardized global skill ontology.
 - **Integration:** These extracted skills are embedded directly into the **skills** array within the W3C-compliant JSON-LD document *before* the SHA-256 hash is generated. This makes the extracted skills a permanently verifiable part of the blockchain record.
- **REQ-F2 (Global Talent Matchmaking Engine):**
 - **Trigger:** A Verifier (Recruiter) accesses the careervault.com/recruit portal.
 - **Action:** The Recruiter inputs desired parameters (e.g., Required Skills: "Python", "AWS").
 - **Ranking Algorithm:** The system queries the PostgreSQL database for candidates who have these exact skills cryptographically verified in their JSON-LD history. Candidates are globally ranked based on:
 - **Skill Match Percentage** (How many requested skills are present in their verified LORs).
 - **Recency Weight** (Newer verified documents carry a higher weight than older documents).
 - **Privacy Constraint:** The system only ranks and displays candidates who have explicitly toggled their profile to "Open to Opportunities," ensuring GDPR compliance and user sovereignty.
- **REQ-F3 (Explainable AI & Transparent Match Justification):**
 - **Trigger:** When a Verifier (Recruiter) views the ranked list of candidates.
 - **Action:** An Explainable AI (XAI) layer computes the feature importance for each candidate's ranking (e.g., using SHapley Additive exPlanations - SHAP).
 - **Output:** The UI provides a "Match Breakdown" for each candidate. Instead of a black-box score, the system visually explains the weight of each factor:

- *Example:* "Total Score 92%: +45% for verified 'Python' skill, +30% for high Document Recency (issued < 6 months ago), +17% for high Issuer Trust Score."
- **Compliance Integration:** The XAI logic ensures the algorithm strictly ignores protected demographic attributes, providing an auditable, transparent decision-making trail that complies with modern algorithmic fairness regulations (e.g., the EU AI Act).

4. Technical Architecture

4.1 Database Schema (Simplified PostgreSQL)

Table Name	Description	Key Columns
organizations	Verified Companies	id, domain, dns_token, is_verified, subscription_tier
users	All Actors	id, email, role (ADMIN, MANAGER, HR, HOLDER)
documents	The LORs	id, holder_id, signer_id, hr_approver_id, json_data, file_hash, status
merkle_roots	The Blockchain Anchors	id, root_hash, tx_hash (Polygon), created_at
access_links	Shared Links	id, document_id, is_active, payment_status

4.2 Security

- **Salting:** All document hashes include a random_salt string stored separately from the hash. If a user requests deletion (GDPR), we delete the JSON + Salt. The on-chain hash is now mathematically "dead" and cannot be linked back to the user.
 - **Magic Links:** Authentication via email tokens (JWT) with short expiry (15 mins) to prevent unauthorized access if an email is forwarded.
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5. User Journey: "The Happy Path"

1. **Sarah (Holder)** requests LOR -> inputs mark@google.com.
 2. **System** checks Google is verified -> Sends link to Mark.
 3. **Mark (Manager)** clicks link -> Signs "Good job, Sarah" -> Submits.
 4. **System** routes to **HR Dashboard**.
 5. **HR (Approver)** sees "Pending LOR" -> Checks dates -> **Approves**.
 6. **System** generates PDF -> Hashes it -> Queues for midnight anchor.
 7. **Sarah** sees "Verified LOR" in dashboard.
 8. **Sarah** pays \$5 -> Gets link careervault.com/verify/abc-123.
 9. **Recruiter** clicks link -> Sees "Verified by Google (HR Approved)."
 10. **System Action:** Behind the scenes, the AI extracts "Project Management" and "Java" from Mark's text and anchors them in Sarah's JSON-LD file.
 11. **Recruiter Search:** A completely different Recruiter searches the CareerVault portal for "Verified Java Developers." Sarah appears at the top of the ranked list because her skills are cryptographically backed by Google.
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